

Project Report

PLACEMENT PORTAL APP V1

1. Student Details

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Course: Modern Application Development – I

Program: IIT Madras BS Degree Program

About me: I am a student in the IIT Madras BS Degree program. I have a good interest in web application development and data-driven technologies. My work focuses on creating useful applications that combine learning, analytics, and user experience.

2. Project Details

Project Title: Placement Portal App V1

Problem Statement:

Traditional placement management systems are inefficient and difficult to manage manually. A digital placement portal is required to automate student job applications, company recruitment, and placement tracking while ensuring security, transparency, and efficient workflow management.

Approach: The application was built using **Flask** as the backend framework with a focus on **role-based access control (RBAC)**. The system handles secure authentication, programmatic database creation, and all.

3. AI/LLM Declaration

ChatGPT (GPT-5.1) were used only for clarity and organizing the structure of the code and only used about 20% of it. The final implementation logic, CRUD operations, validation, and integration were done manually. The frontend UI was created with Bootstrap, with only a little AI help in planning the content. The project is original and not copied from external sources.

4. Technologies and Frameworks Uses

Technology	Purpose
Flask	Backend routing & controllers
SQLite	Lightweight relational database
Jinja2	HTML template rendering
Bootstrap 5	Responsive frontend UI
Chart.js	Data visualization for admin dashboard
Flask-Login	Authentication & session management

5. System Architecture

Frontend

- Created using HTML, CSS, and Bootstrap
- Responsive on mobile
- Uses Jinja2 for dynamic pages
- Simple JS validations

Backend (Flask)

- Routing for all modules
- Login and logout system
- Role-based access control

Database (SQLite)

Tables used:

- Users – Stores login credentials and roles
- StudentProfile – Stores student academic and profile details
- CompanyProfile – Stores company information
 - Jobs – Stores posted job details
- Applications – Stores job applications submitted by students
- Placements – Stores placement records of selected students

6. DATABASE SCHEMA

Users Table

- id (Primary Key)
- username
- email
- password
- role

Jobs Table

- id (Primary Key)
- company_id (Foreign Key → CompanyProfile.id)
 - title
- description
- skills_required
- is_approved
 - location
- salary_range
- job_type

Application Table

- id (Primary Key)
- job_id (Foreign Key → Jobs.id)
- student_id (Foreign Key → StudentProfile.id)
 - status
 - applied_at
 - updated_at
 - remarks

Placements Table

- id (Primary Key)
- application_id (Foreign Key → Applications.id)
 - placed_at
 - package

Student Profile

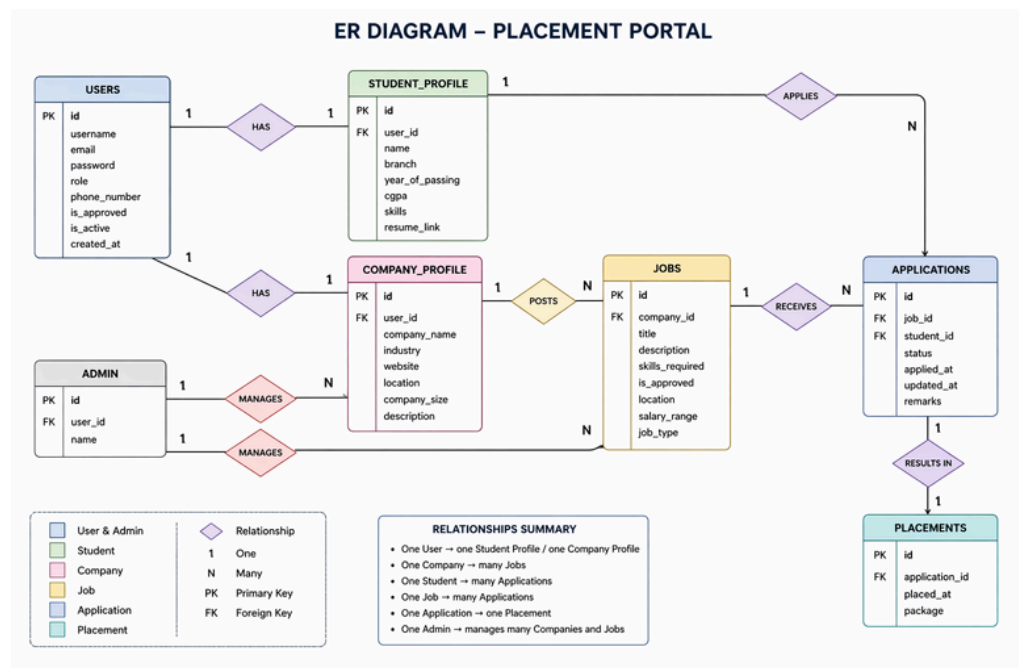
- id (Primary Key)
- user_id (Foreign Key → Users.id)
- name
- branch
- year_of_passing
- cgpa

Company Profile

- id (primary key)
- user_id (Foreign Key → Users.id)
- company_name
- industry
- location

ER Diagram Summary

- One user → one student profile / one company profile
- One company → many job postings
- One student → many job applications
- One job → many applications
- One application → one placement record
- One company → many applicants
- One admin → manages many companies and jobs



7. CORE FEATURES IMPLEMENTED

- User Authentication and Login System
 - Role-Based Access Control (Admin, Company, Student)
 - Student Registration with Resume Upload
 - Company Registration with Admin Approval
 - Job Posting and Management System
 - Job Search and Filtering Functionality
 - Job Application Submission and Tracking
 - Duplicate Application Prevention
 - Placement Status Management
 - Admin Dashboard for Monitoring and Approval
 - Company Dashboard for Applicant Management
 - Student Dashboard for Application Tracking
 - Chart-Based Data Visualization
 - REST API Integration for JSON Data
 - Frontend and Backend Validation
 - Responsive User Interface using Bootstrap
 - Secure Session Management using Flask-Login
 - Password Hashing and Protected Routes
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8. API ENDPOINTS

GET /api/students – Retrieve list of all student profiles

GET /api/companies – Retrieve list of all company profiles

GET /api/jobs – Retrieve all approved job postings

GET /api/applications – Retrieve all submitted applications

POST /api/job – Create a new job posting

PUT /api/application/<id> – Update status of a specific application

DELETE /api/job/<id> – Remove a job posting from the system

9. SECURITY & VALIDATION

Frontend

- Required fields
- Email check
- Password length

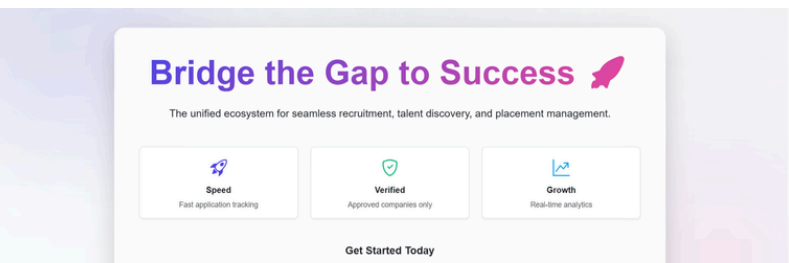
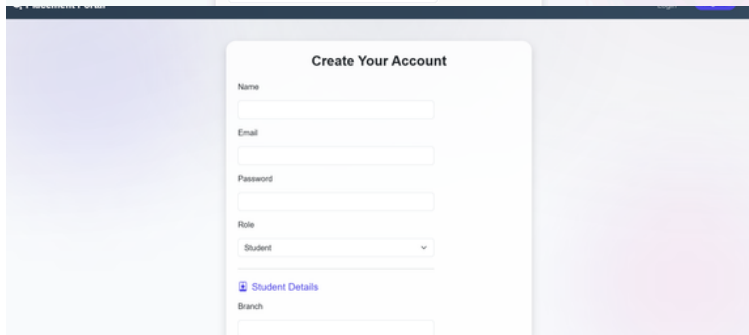
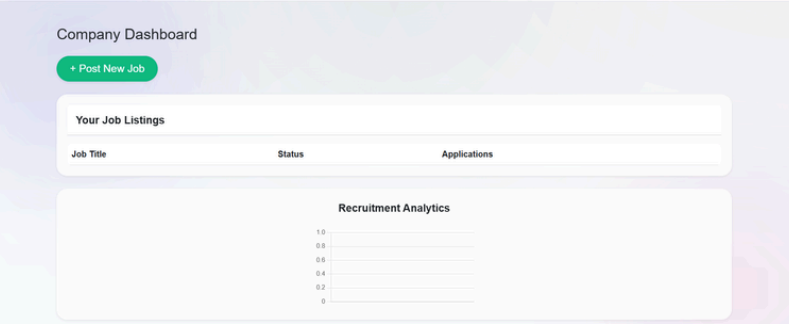
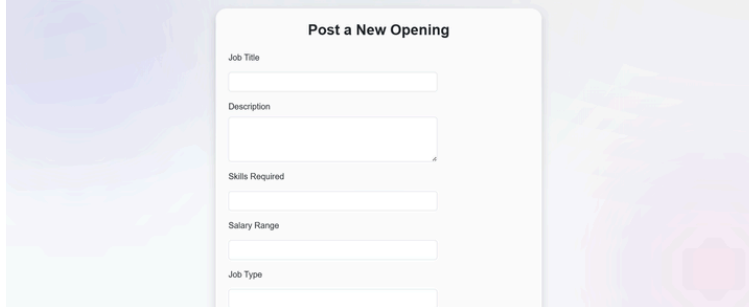
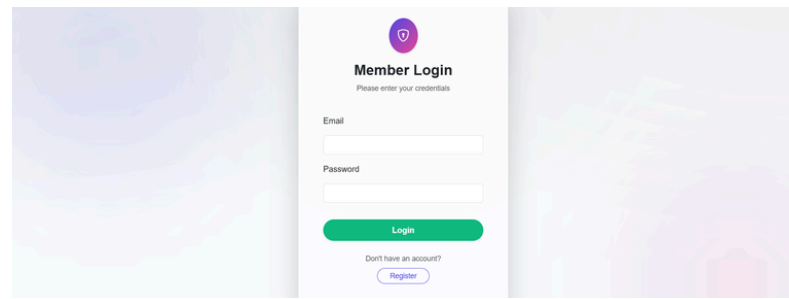
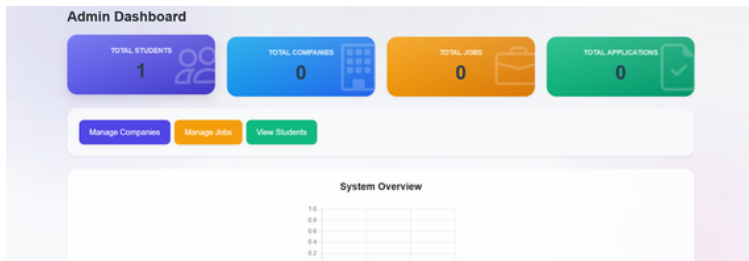
Backend

- Prevent duplicate emails
- Prevent double booking
- Protect routes using roles

Password Security

- Password hashing
 - Flask-Login sessions
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Screenshots of Web App



10. CONCLUSION

The Placement Portal Application provides an efficient and secure solution for managing the placement process digitally. It simplifies job posting, application tracking, and placement management for students, companies, and administrators while improving transparency and reducing manual effort.

11.HMS Project Video Link

[PPA V1 Project Video Link](#)